

AUTONOMOUS MULTI-STRATUM ANOMALY CORRELATION ENGINE [AMACE v2.0]

Domain-Agnostic, Zero-Configuration Causal Discovery

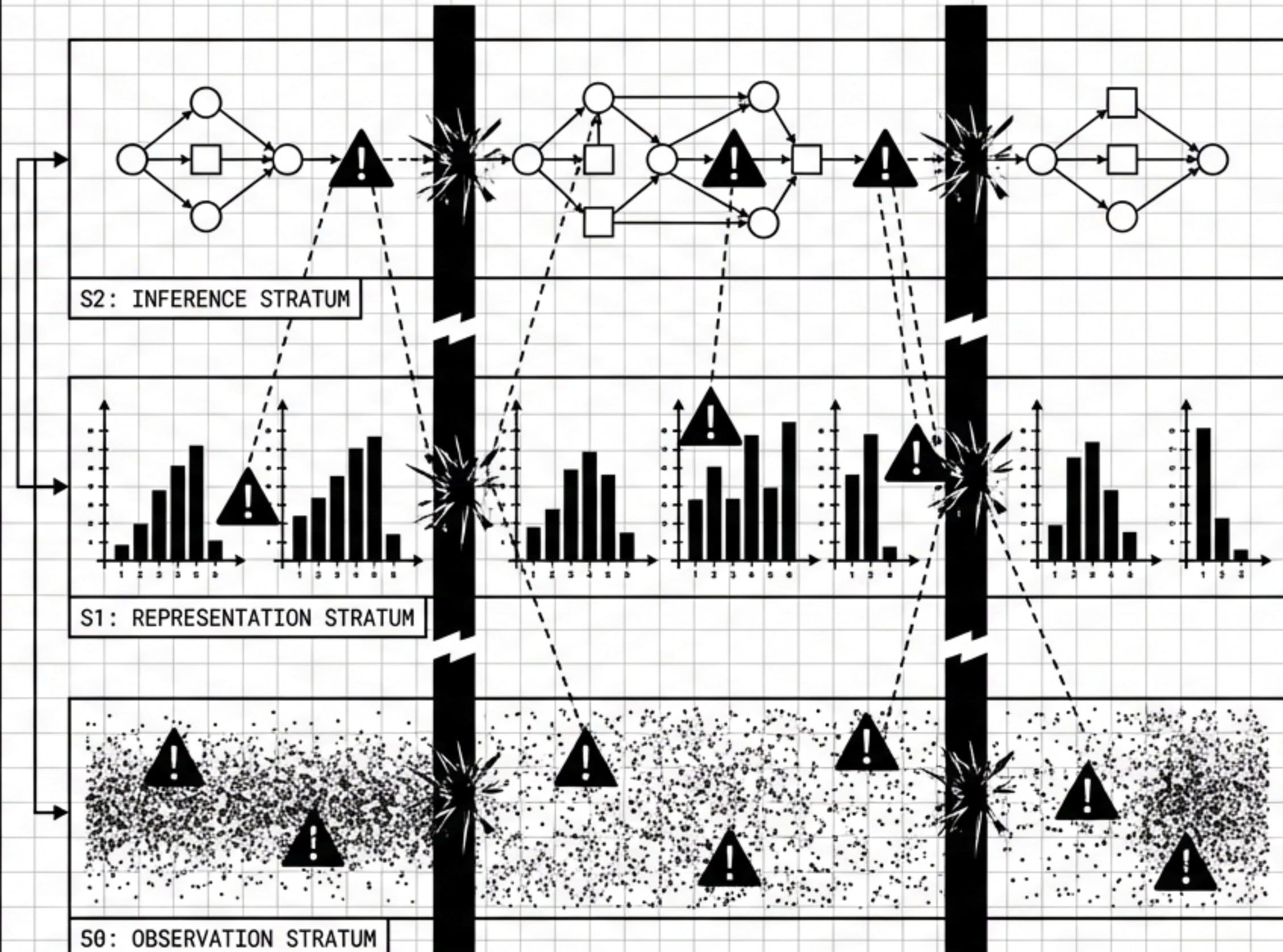
SYSTEM ARCHITECTURE & PIPELINE SCHEMATIC

THE CROSS-STRATUM BLIND SPOT

MODERN SYSTEMS OBSERVE
ANOMALIES IN ISOLATION
ACROSS THREE FUNDAMENTAL
STRATA (S0, S1, S2).

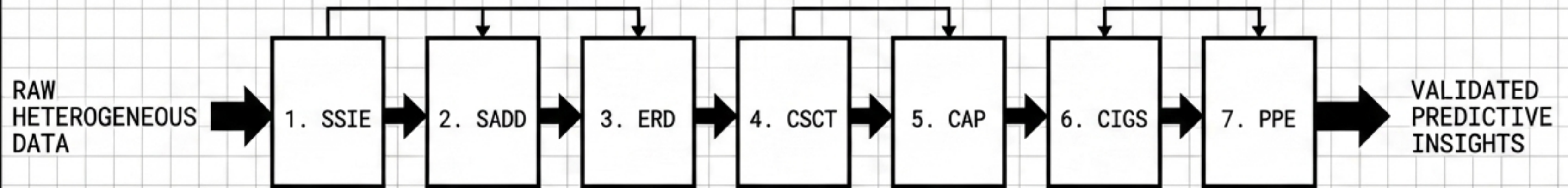
TRADITIONAL TOOLS CANNOT
RESOLVE ENTITY IDENTIFIERS
ACROSS THESE HETEROGENEOUS
BOUNDARIES.

THE COMBINATORIAL EXPLOSION:
CORRELATING N ANOMALIES
ACROSS 3 STRATA WITH D
DIMENSIONS AND K CARDINALITY
CREATES AN $O(N^2 * K^D)$
SEARCH SPACE.
EXCEEDS 10^{18} COMBINATIONS.



ARCHITECTURE OVERVIEW: THE AMACE PIPELINE

Reducing computational complexity from $O(N^2 * K^D)$ to $O(N * \log(N) * k * \log(D))$ through seven strictly bounded processing modules.

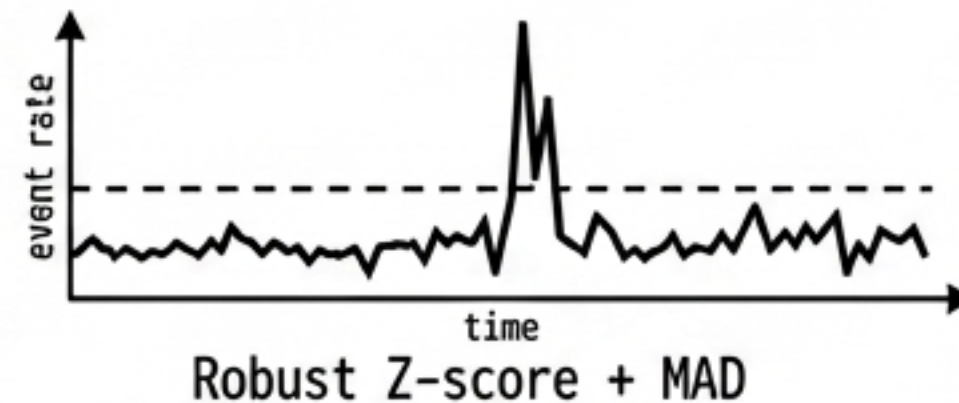


MODULE 2 | STRATUM-ADAPTIVE DEVIATION DETECTOR (SADD)

ROLE: Detect anomalous deviations within each stratum using localized statistical characteristics, outputting a standardized signature.

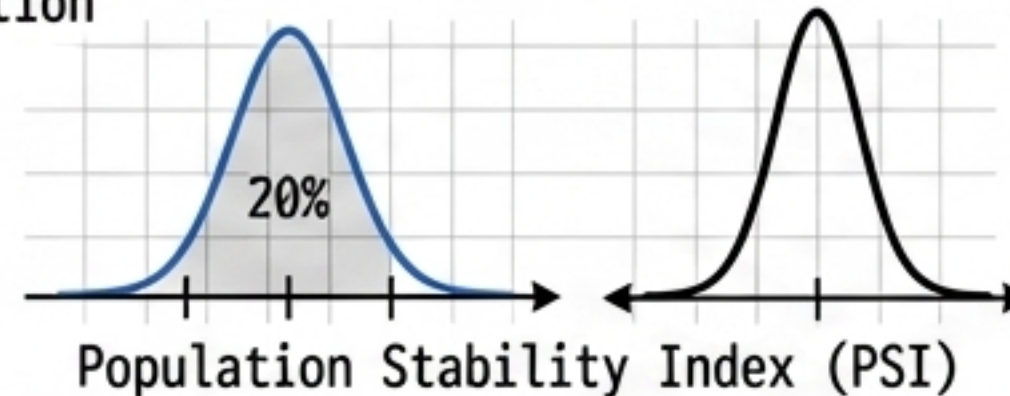
S0 Data

S0 Observation



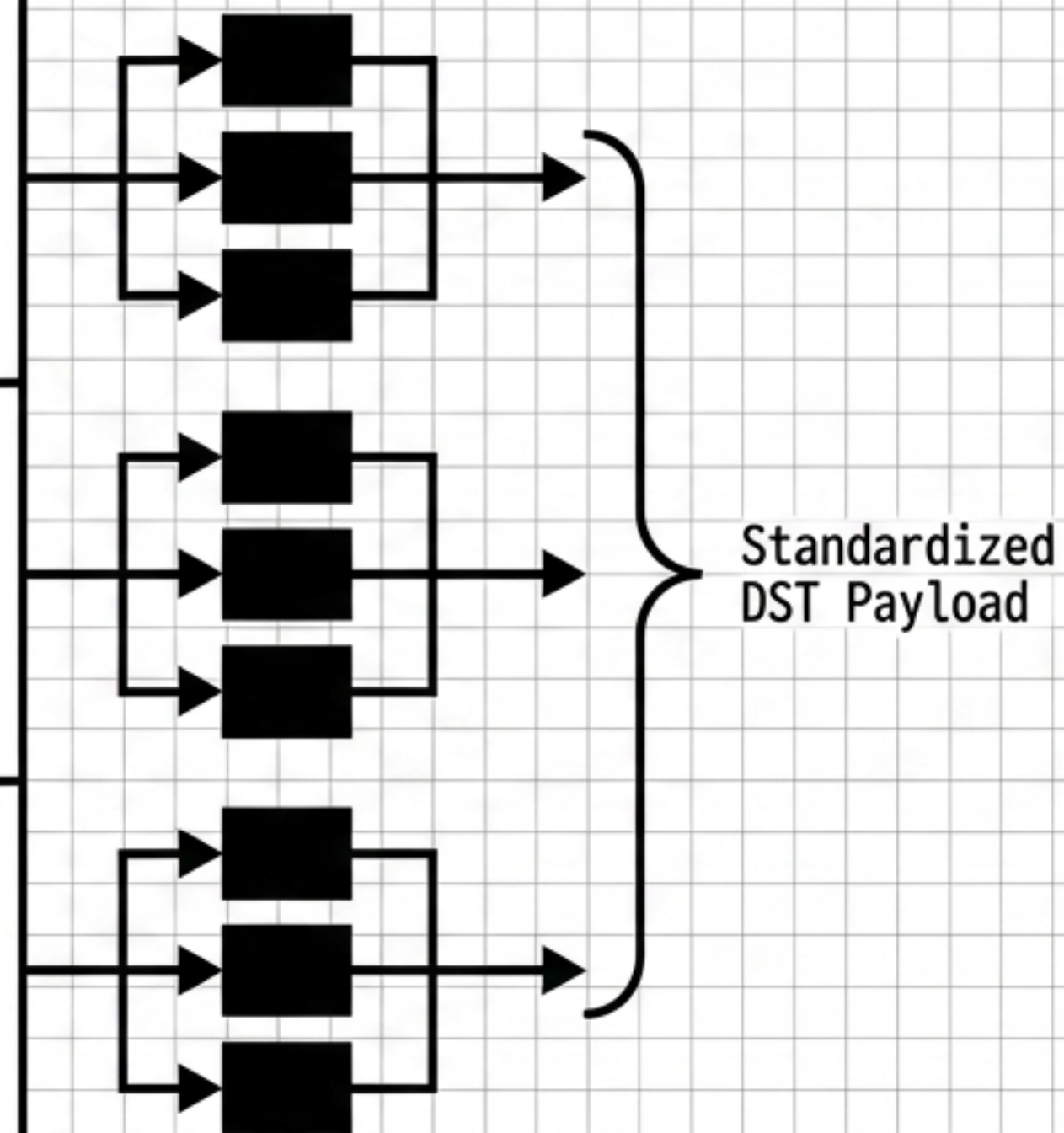
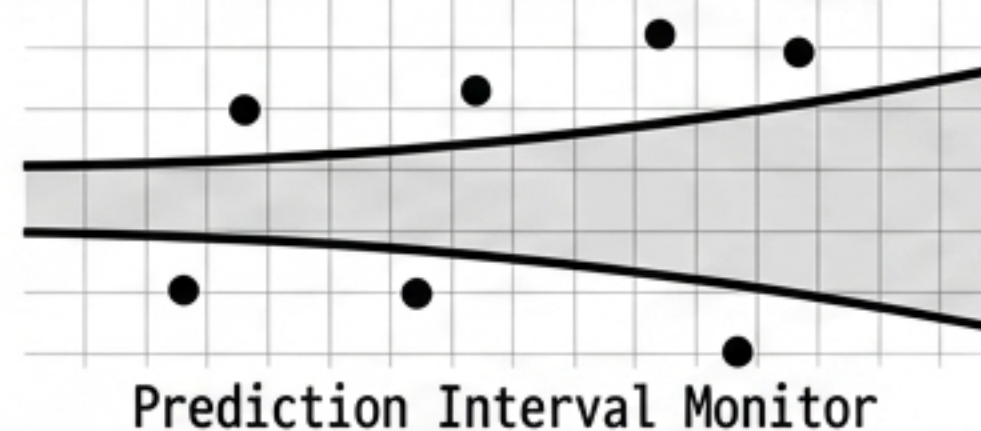
S1 Data

S1 Representation



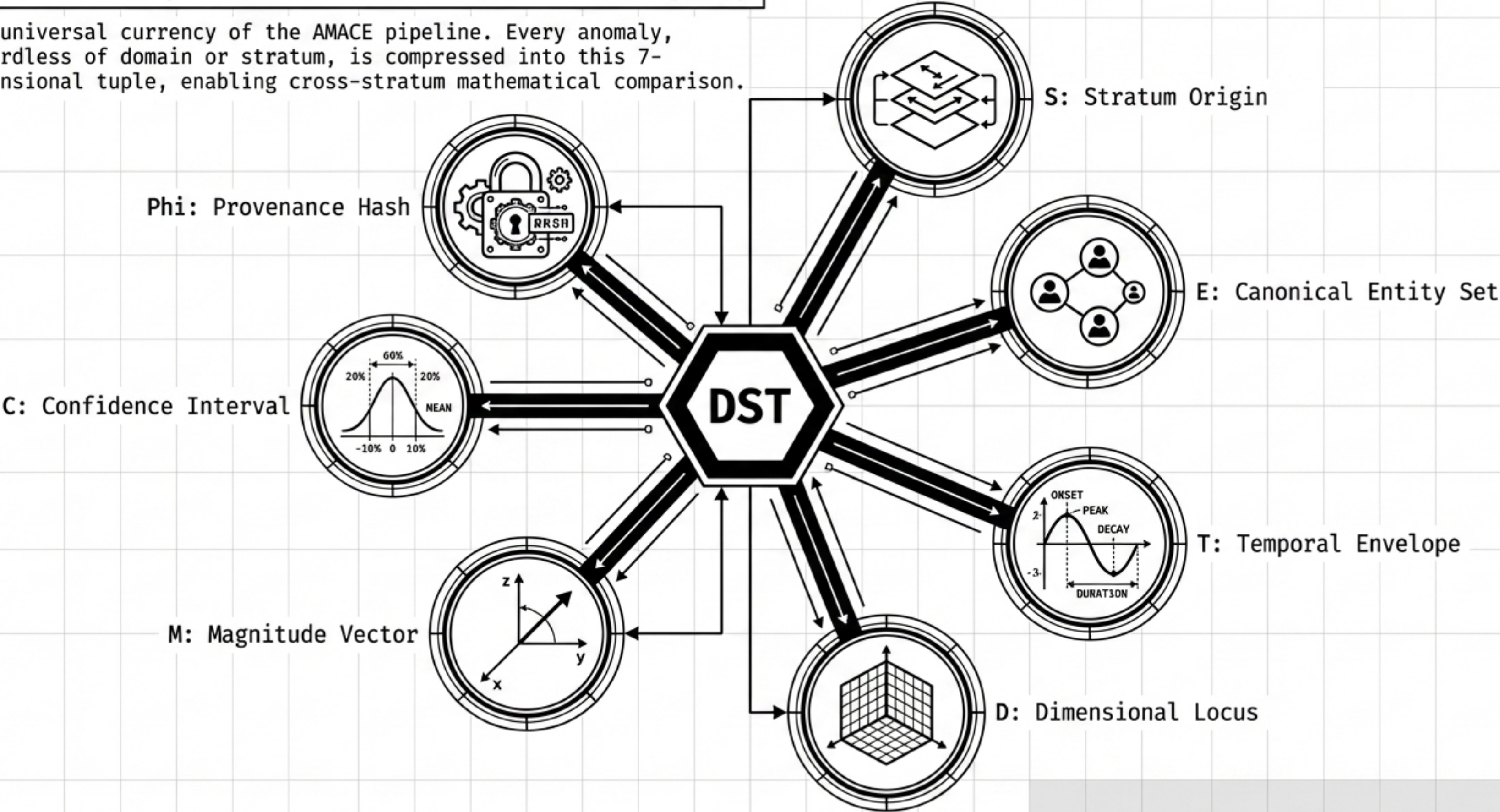
S2 Data

S2 Inference



THE PAYLOAD | DEVIATION SIGNATURE TENSOR (DST)

The universal currency of the AMACE pipeline. Every anomaly, regardless of domain or stratum, is compressed into this 7-dimensional tuple, enabling cross-stratum mathematical comparison.



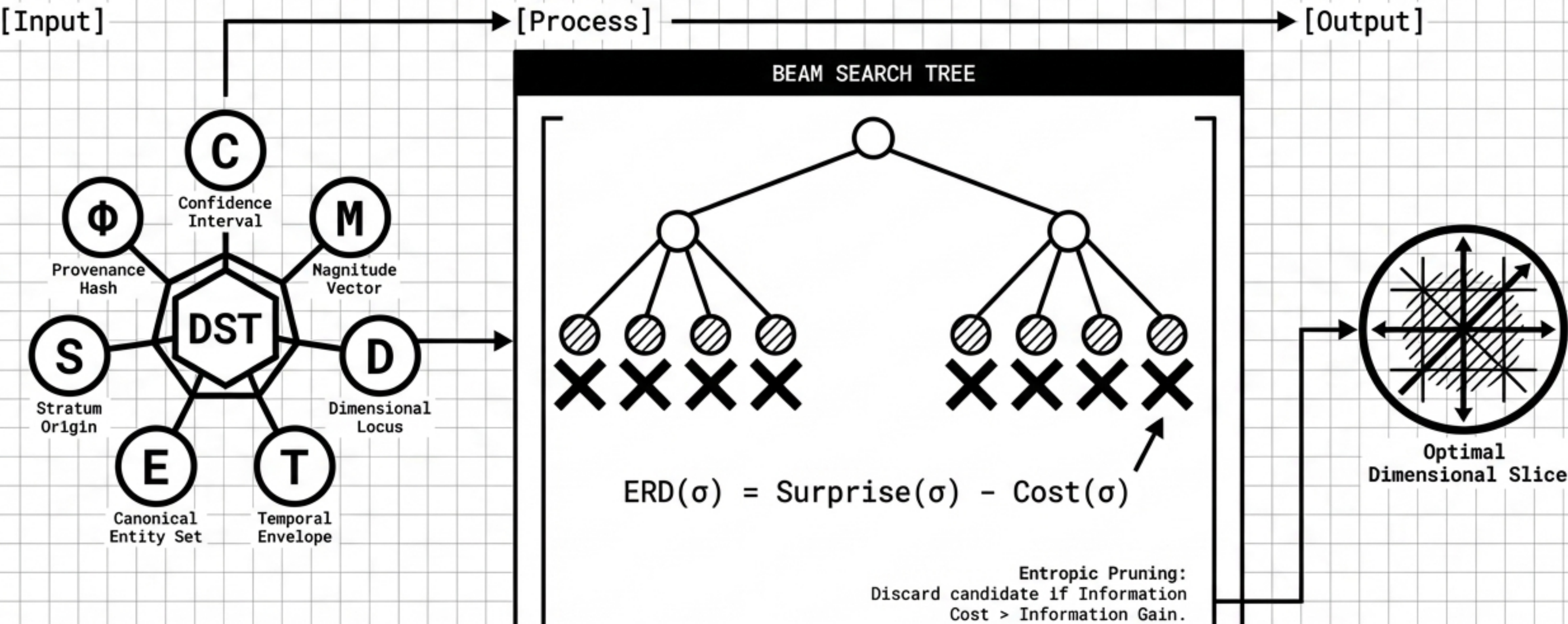
MODULE 3 | ENTROPIC RELEVANCE DECOMPOSITION (ERD)

ROLE:

Identify the specific sub-population where the anomaly is concentrated.

TECHNICAL IMPACT:

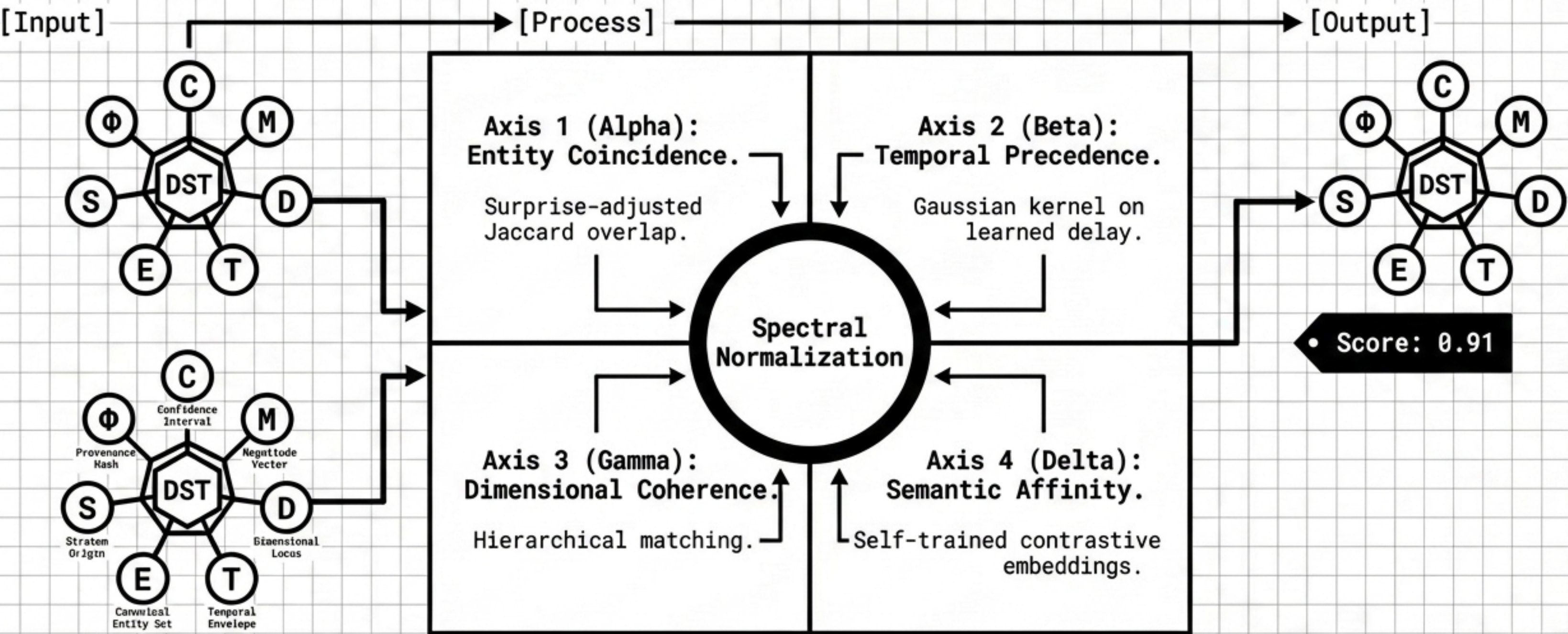
Exponentially reduces search space from $O(K^D)$ (approx. 10^{16}) to $O(w * D * K_{\max})$ (approx. 40,000 evaluations).



MODULE 4 | CROSS-STRATUM CAUSAL TENSOR (CSCT)

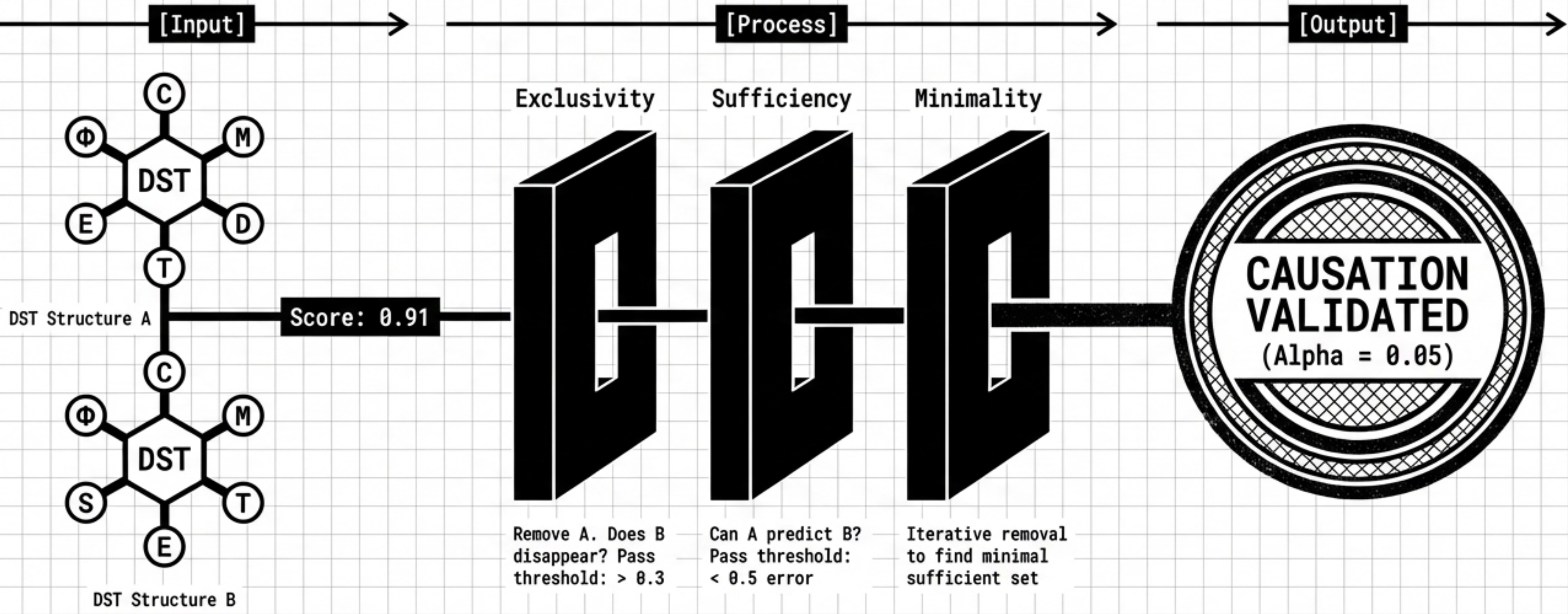
ROLE

Compute a normalized, four-axis similarity score between cross-stratum deviations to detect shared causal origins.



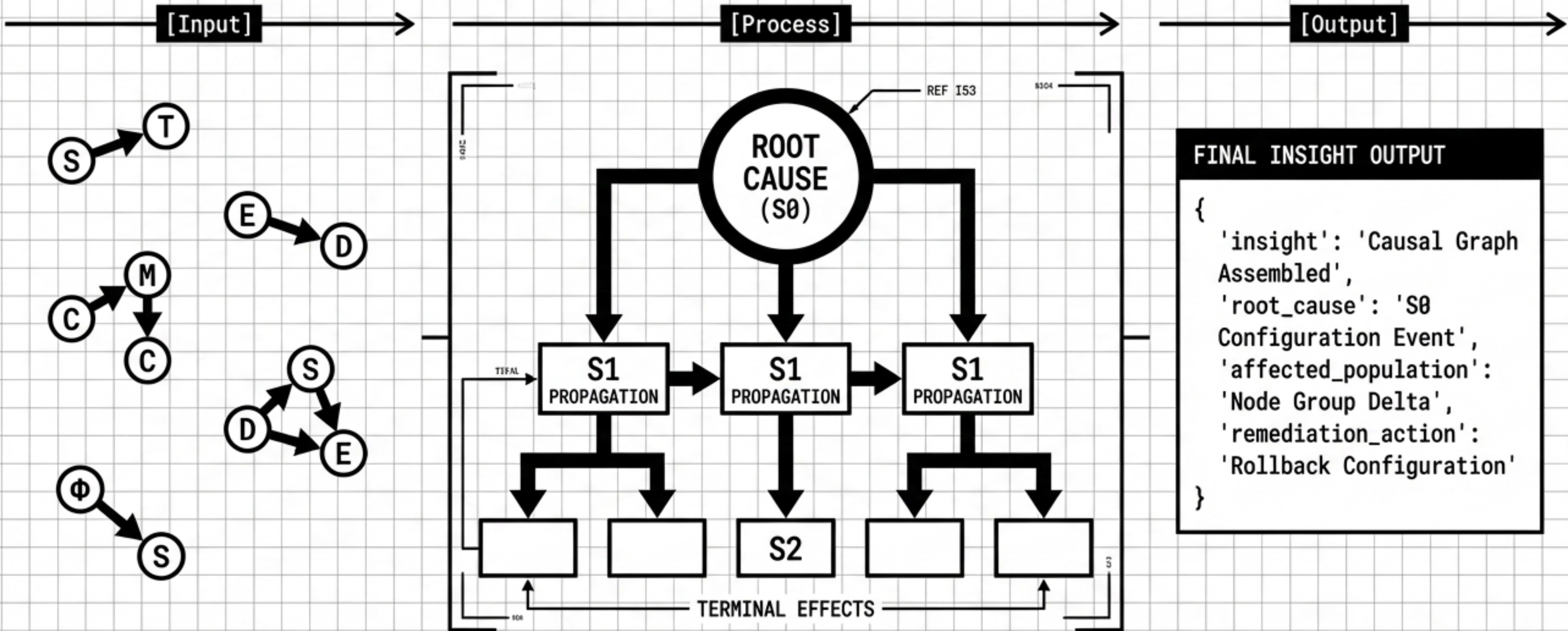
MODULE 5 | COMPUTATIONAL ABLATION PROTOCOL (CAP)

ROLE: Distinguish genuine causal relationships from coincidental co-occurrence through automated counterfactual testing. Bounds Type I error rate at 95% confidence.



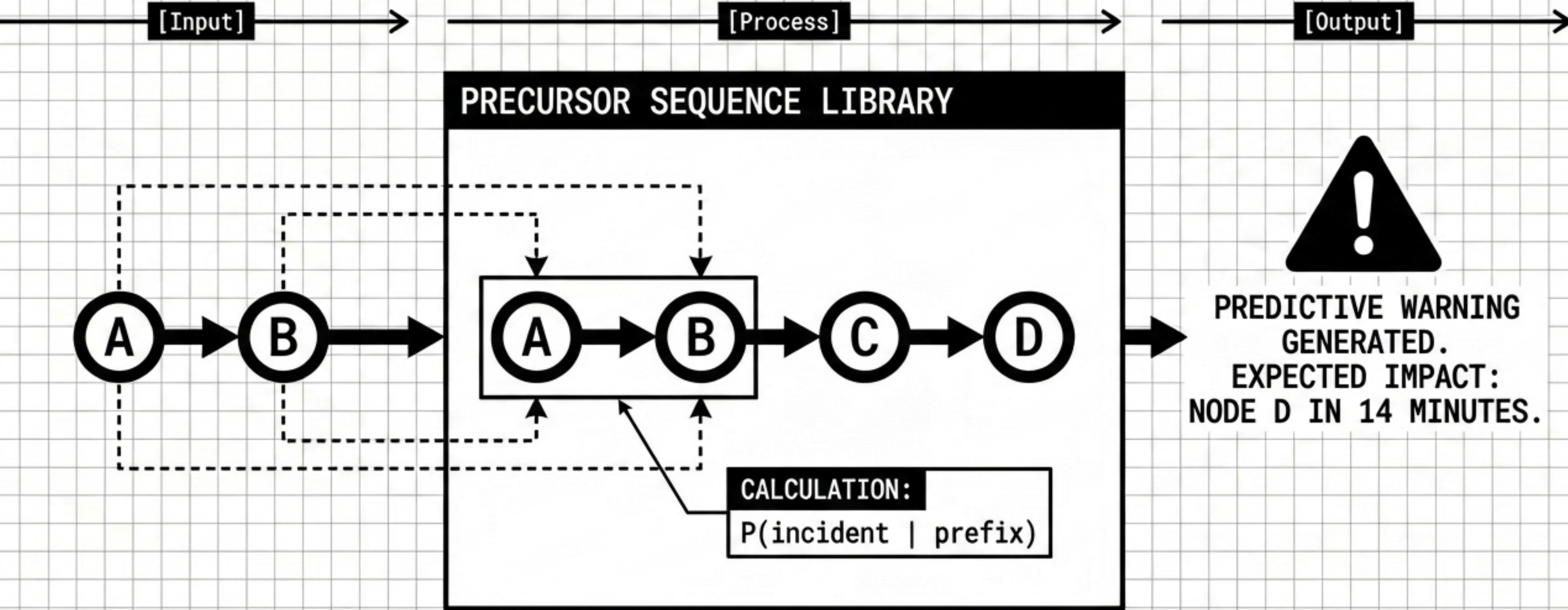
MODULE 6 | CAUSAL INSIGHT GRAPH SYNTHESIZER (CIGS)

ROLE: Assemble validated links into a Directed Acyclic Graph (DAG) exposing root cause, blast radius, and remediation steps.



MODULE 7 | PREDICTIVE PRECURSOR ENGINE (PPE)

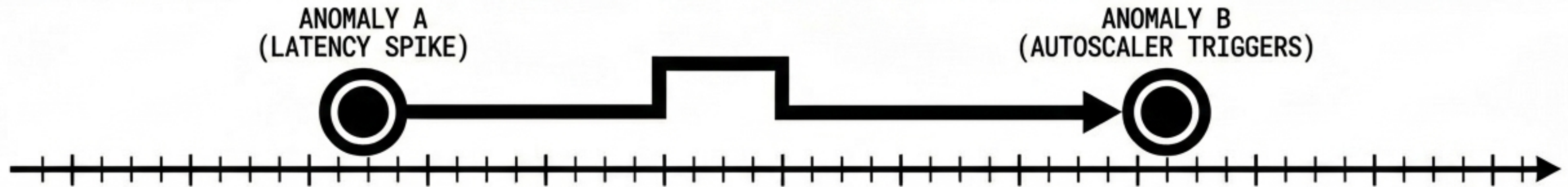
ROLE: Compare real-time causal chains against historical insight graphs to generate forward-looking warnings before terminal impact.



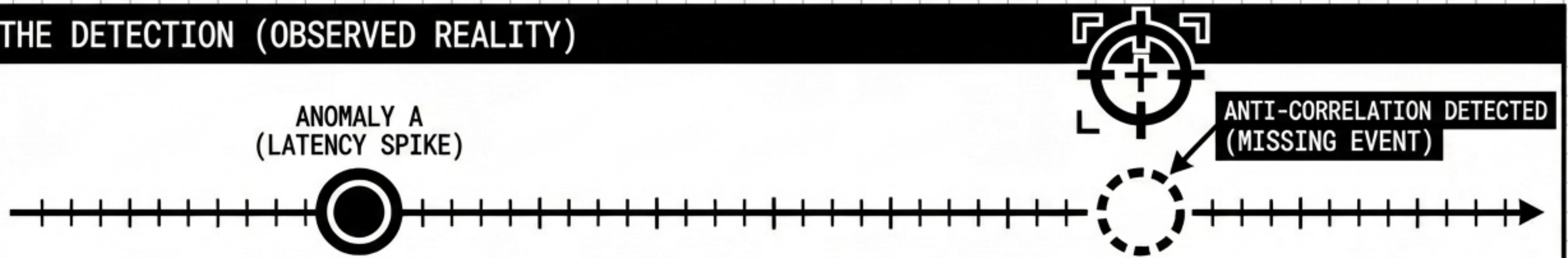
THE ANTI-CORRELATION CAPABILITY

Traditional systems only alert on what fires. AMACE computationally models expected causal chains and alerts when an expected downstream correlation is missing, revealing latent misconfigurations and silent failures.

THE LEARNED BASELINE (EXPECTED BEHAVIOR)



THE DETECTION (OBSERVED REALITY)

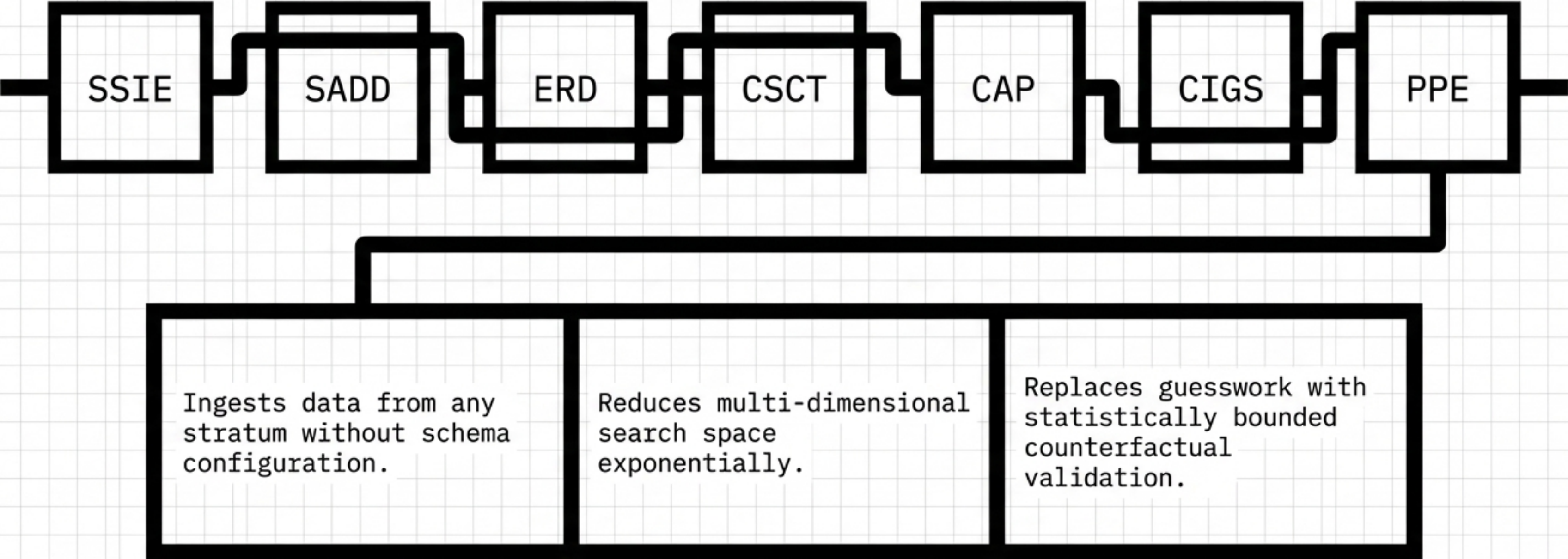


ARCHITECTURAL DIFFERENTIATION MATRIX

A measurable technical improvement to the functioning of the computer system itself.

DIMENSION	PRIOR ART / AIOPS	AMACE ARCHITECTURE
COMPUTATIONAL COMPLEXITY	$O(N^2 * K^D)$ (Exhaustive search limit)	$O(N * \log(N) * k * \log(D))$ (Entropic pruning)
CONFIGURATION & SCHEMA	Manual mapping, predefined rules	Zero-configuration Stratum Schema Inference Engine (SSIE)
ANOMALY CURRENCY	Heterogeneous, domain-specific alerts	Universal 7-dimensional Deviation Signature Tensor (DST)
CAUSALITY VALIDATION	Co-occurrence clustering (correlation only)	Computational Ablation Protocol (CAP) with 3 counterfactual tests

FROM BLIND SPOTS TO PROVABLE CAUSALITY



AUTONOMOUS, MULTI-STRATUM ANOMALY CORRELATION—PROVABLY WITHIN SUB-QUADRATIC COMPUTATIONAL COMPLEXITY.